PRODUCT REQUIREMENTS DOCUMENT

Saudi Legal AI Assistant

A source-grounded legal RAG platform with monitored regulatory updates

PRODUCT POSITIONING

An Arabic-first, citation-driven legal information assistant that indexes a curated collection of official Saudi legislation, provides article-level answers, monitors official publications for changes, and maintains version-aware retrieval. This is an MVP for AI-engineering hiring assessment—not a production legal advisory service.

DOCUMENT CONTROL	DETAIL
Document status	Ready for implementation / technical assessment
Version	1.0
Date	12 October 2026
Intended audience	Hiring panel, AI engineer, backend developer, product reviewer
MVP duration	14 working days (planning assumption)
Primary language	Arabic for user questions and legal source text; English PRD
Scope	10–20 curated official documents across 3 domains

Success principle

Trustworthy citations, legal effective-date handling, and reproducible update workflows matter more than corpus size or a polished chatbot animation.

Document map

SECTION	COVERAGE
01–04	Executive brief · Problem · Goals · Users and journeys
05–08	Scope · Official sources · Functional requirements · Update lifecycle
09–12	RAG design · Architecture · Data model · APIs and UI
13–16	Security · Nonfunctional requirements · Evaluation · Delivery plan
17–20	Demo · Risks · Hiring rubric · References and glossary

READING NOTE

All target metrics and sprint dates below are proposed acceptance criteria, not measured results. Source availability, access conditions, and legal validity must be independently confirmed during implementation.

01. Executive summary

Saudi Legal AI Assistant is an MVP web application that answers questions about selected Saudi regulations using retrieval-augmented generation (RAG). Unlike a generic chatbot, each substantive legal statement must be grounded in a retrieved official source with a clickable citation to the underlying law or regulation and, where available, its article number and applicable version.

A companion ingestion and update workflow acquires approved legal documents, creates article-aware searchable records, monitors the official Umm Al-Qura publication feed for possible changes, compares document versions, and queues updates for reviewer approval. The RAG engine distinguishes publication date from effective date and never treats an unreviewed update as authoritative.

The product is intentionally scoped as a hiring exercise. It demonstrates end-to-end AI engineering, retrieval design, evaluation discipline, source governance, backend APIs, data modeling, and practical deployment without claiming complete coverage of Saudi law.

02. Problem, opportunity, and hypothesis

2.1 Problem statement

Saudi laws and regulations are distributed across official repositories and issuing authorities. Users may struggle to find the controlling provision, distinguish current rules from superseded text, or learn whether a newly published change is already effective. Generic LLM answers can be confident but uncited or temporally incorrect.

2.2 Product hypothesis

For a narrowly defined corpus, article-aware hybrid retrieval, strict evidence-backed answer generation, and a version-controlled change pipeline can produce more auditable legal information answers than free-form prompting alone.

2.3 Outcomes

- Users can ask Arabic legal-information questions and inspect the authoritative legal text used in the answer.
- Administrators can upload, inspect, approve, and remove document versions without manual database intervention.
- The platform can detect potential new publications, verify what changed, and publish reviewed updates safely.
- An evaluator can reproduce tests, review logs, and run a documented live demo.

03. Goals, KPIs, and guardrails

CATEGORY	MEASURABLE TARGET / DECISION RULE
Corpus coverage	10–20 legally sourced documents in 3 selected domains, with article IDs where applicable.
Citation correctness	≥95% of citations in 50 test questions point to the correct source and passage (human-graded).
Answer grounding	≥90% of answerable test questions have no unsupported material legal claims (human-graded).
Abstention	100% of clearly out-of-corpus test items must not invent a legal answer.
Temporal validity	100% of predefined effective-date regression scenarios identify the correct version/status.
Update workflow	One complete simulated amendment: discovery → review → approval → re-index → changed answer.
Performance	p95 ≤8 seconds for normal chat requests on the test environment (excluding cold starts).
Observability	Each answer carries request ID, retrieved article/version IDs, processing timestamps and error status.

HARD GUARDRAIL

A good answer with a wrong statutory citation fails. The system must explicitly abstain or qualify whenever the evidence is missing, conflicting, not effective, or outside the indexed coverage. The application must never represent itself as providing professional legal advice.

04. Users and core journeys

PERSONA	JOB TO BE DONE	REQUIRED CAPABILITY
End user	Find what a selected Saudi law says about a specific issue.	Ask in Arabic; open exact source; see effective-date note.
Admin / reviewer	Manage source documents and decide if updates are trustworthy.	Upload, compare, approve, reject and roll back versions.
Hiring evaluator	Verify the engineering system rather than trust screenshots.	Run reproducible tests, inspect logs and follow README setup.

4.1 Primary user journey

1. User enters a question in Arabic, optionally selects domain and “as-of” date.
2. System retrieves the most relevant article versions valid for the requested date.
3. System responds in Arabic with a concise explanation, an uncertainty or scope note where appropriate, and clickable citations.
4. User opens the source document and can inspect the supporting passage.

4.2 Update reviewer journey

1. Scheduled monitor records a new item or changed document fingerprint from an approved government source.
2. An update candidate appears with original URL, publication metadata, proposed classification and extracted content.
3. Reviewer compares old and new articles; checks legal status, source and effective date.
4. Reviewer approves or rejects. Approved changes become searchable in the appropriate temporal view; superseded material remains auditable.

05. MVP scope and release boundaries

5.1 In scope (P0)

- Arabic question-answering against a curated, official legal corpus with article-level citations and source URLs.
- PDF and HTML/text ingestion for authenticated admins; parsing, normalization and structured metadata.
- Hybrid retrieval (keyword + vector), source filtering, basic relevance scoring and answer abstention.
- Scheduled monitoring of selected official feeds/pages, including Umm Al-Qura RSS for decisions and regulations.
- Update candidate detection, de-duplication, manual approval, immutable version history and re-indexing.
- As-of-date filtering and clear differentiation of published, pending-effective, current and superseded rules.
- Basic responsive web chat, admin review panel, logs, evaluation suite, Docker deployment and README.

5.2 Stretch (P1)

- English-language questions with Arabic controlling-source citations; historical comparison UI; OCR for scanned PDFs; email alerts; query analytics; semantic re-ranking service; exportable answer as PDF.

5.3 Explicitly out of scope

- Indexing all Saudi legislation or guaranteeing complete jurisdiction-wide coverage.
- Unsupervised interpretation of every government announcement as a legally effective amendment.
- Automated legal opinions, dispute outcome prediction, legal representation or filing submissions.
- Authentication through Saudi national identity providers, billing, mobile apps, complex RBAC hierarchies.
- Court judgments/case-law research unless separately approved as a future corpus.

06. Official sources and data governance

SOURCE	PROPOSED USE	IMPLEMENTATION NOTE
Bureau of Experts – Saudi Laws [S1]	Primary searchable legal corpus; legal document reference.	Review access and reuse permissions; validate canonical URLs.
Umm Al-Qura [S2]	Official publication/notice reference.	Publication is not the same as effective date.
Umm Al-Qura RSS [S3]	Discover candidate regulatory updates.	Feed item is a detection signal, not final validation.
HRSD – Labor Law [S4]	Initial labor domain corpus.	Prefer the currently approved text and verify amendments.
Ministry of Commerce [S5]	Initial corporate/commercial domain corpus.	Retain originating authority and canonical citations.
SDAIA – PDPL knowledge center [S6]	Initial data-protection domain corpus.	Distinguish law, implementing regulations and guidance.

The implementation must use a configurable source allowlist, observed source URLs, polite request frequency, error handling, and a source-licensing/access review. Do not assume that crawling every government site is permitted or that a public website offers a supported API. Prefer published feeds and permitted downloads; where access is restricted, perform manual admin upload.

6.1 Corpus selection for MVP

Suggested pilot domains: (1) employment/labor; (2) companies/commercial rules; (3) personal data protection. Select documents with stable official access, clearly named provisions, and enough testable amendments or version metadata. Keep a signed-off corpus manifest with document title, source authority, source URL, extraction date, review status, edition and checksum.

6.2 Source priority and conflict resolution

- Use authoritative Arabic legal text as the controlling basis; translations and ministry explanatory pages are supplemental, not replacements for governing text.

- Maintain publication source and consolidated-text source separately when they serve different purposes.
- If two official sources appear inconsistent, surface a warning and queue human review rather than silently select one.
- Published notices, draft consultations, news summaries, executive regulations and binding legal instruments receive distinct document-type tags.

07. Functional requirements

Priorities: P0 = release blocker; P1 = enhancement. Each requirement includes testable acceptance criteria. IDs are stable references for issues, test cases and demo steps.

7.1 Ingestion and source management

ID	REQUIREMENT (P0)	ACCEPTANCE CRITERIA
FR-01	Admin uploads approved PDF/text or registers an approved HTML URL.	File/source validated; job ID returned; unauthorized users blocked.
FR-02	Parser extracts readable text and legal structure (title, chapters, article numbers).	≥95% accurate article ID extraction on the curated test corpus.
FR-03	Arabic-aware normalization for search while storing exact original text.	Search variants match; original punctuation/numbering retrievable.
FR-04	Store provenance, issue date, publication date, effective date, authority and checksum.	Missing critical metadata marked “needs review”; never inferred as fact.
FR-05	Idempotent ingestion and duplicate detection.	Same file hash and same version do not create duplicate active chunks.

7.2 Search, answer generation and citations

ID	REQUIREMENT (P0)	ACCEPTANCE CRITERIA
FR-06	Accept Arabic natural-language queries and optional domain/as-of date filters.	Validated inputs; filter values appear in request trace.
FR-07	Use hybrid lexical + embedding retrieval, preserving article boundaries.	Top-k results expose article IDs, rank and scores.
FR-08	Generate answers only from validated retrieved evidence.	No citation-free statutory claims in answerable gold set.
FR-09	Cite law title, article number, source URL and applicable version when available.	Citation opens intended canonical resource and supports claim.
FR-10	Abstain for missing, conflicting or weak evidence.	All out-of-corpus tests produce a clear limitation statement.
FR-11	Respect legal effective-date state in retrieval and response.	Published-but-not-effective material excluded from default current answer.

7.3 Monitoring, review, and versioning

ID	REQUIREMENT (P0)	ACCEPTANCE CRITERIA
FR-12	Poll official RSS/feed or approved source pages at scheduled intervals.	Run history visible; errors retried and logged.
FR-13	Identify candidates via canonical URL, item ID and content hash.	Repeat feed reads do not create duplicate candidates.
FR-14	Classify candidate as new law, amendment, regulation, notice or irrelevant.	Human reviewer can override classification.

ID	REQUIREMENT (P0)	ACCEPTANCE CRITERIA
FR-15	Compare text/article changes and identify potentially affected instruments.	Reviewer sees before/after snippet and affected article IDs.
FR-16	Require review and approval before material becomes authoritative.	Unapproved content never enters user-facing authoritative retrieval.
FR-17	Keep immutable versions and allow safe rollback.	Historical versions searchable with requested as-of date.
FR-18	Re-index approved, time-valid content and invalidate stale search data.	Version flip changes test answer and leaves audit record.

7.4 Administration and operations

ID	REQUIREMENT (P0)	ACCEPTANCE CRITERIA
FR-19	Role-based login: public read-only user and admin/reviewer.	Only admins can upload, approve, modify or roll back.
FR-20	Dashboard displays corpus, update queue, jobs and error state.	Reviewer can inspect per-run state and approve/reject.
FR-21	Persist structured request/response traces excluding unnecessary personal content.	Trace links answer ↔ retrieved citations ↔ document version.
FR-22	Expose health and readiness checks.	Health endpoint reflects database and worker availability.
FR-23	Ship automated tests, fixture corpus and evaluation report.	Clean-environment evaluator can reproduce stated results.

08. Legal update lifecycle and time validity

8.1 Discovery-to-publication workflow

STEP	PROCESS	OUTPUT
1. Detect	Poll approved official feed/page, record item URL, title and timestamp.	Raw candidate + source trace.
2. De-duplicate	Normalize identifier and compare content fingerprints.	New/unchanged classification.
3. Acquire	Fetch permissible official text or request admin upload.	Immutable source snapshot.
4. Extract & align	Parse Arabic articles, map amendment to affected instrument.	Structured proposed diff.
5. Verify	Review source status, authority, publication/effective dates, affected text.	Approved or rejected review event.
6. Version & index	Store new version; index with effective-time metadata.	New searchable version + preserved history.
7. Observe	Run regression queries, record result and alert on failure.	Audit trail and rollback option.

8.2 Legal-status state machine

Candidate → Retrieved → Parsed → Needs Review → Approved → Scheduled (future effective date) → Active → Superseded / Repealed. Rejected and Failed are terminal or retryable side paths. “Approved” means the ingestion was verified; it does not by itself mean the legal rule is currently effective.

8.3 Temporal retrieval semantics

- Default question ("What is the rule today?"): retrieve documents effective on the query date and not repealed or superseded on that date.
- "What changed?" question: return a before/after summary only if both versions are verified and cite each separately.
- Future-effective laws: may be discussed as forthcoming rules with an explicit effective date, never conflated with rules currently in force.
- Unknown effective dates: trigger review; do not silently substitute publication date as effective date.
- Maintain both observed_at (system timestamp) and valid_from / valid_to (legal validity) because system knowledge time differs from legal effect time.

UPDATE SAFETY CONTROL

MVP must use review-before-publish. Feed polling is automatic; promotion of a legal change to authoritative answers is conditional on approval and confirmed legal metadata. A simulated amendment is acceptable for the hiring demo if clearly labeled as test data.

09. RAG design and answer policy

9.1 Retrieval pipeline

1. Preprocess query: detect Arabic/English, normalize for retrieval, extract optional statute name, article hint and as-of date.
2. Apply domain, source authority, document type, review state and legal effective-time filters before ranking.
3. Retrieve using pgvector embeddings plus PostgreSQL full-text / trigram or a comparable Arabic lexical index.
4. Combine rankings using reciprocal rank fusion (RRF) or an empirically justified method; optional re-ranker is P1.
5. Retain complete article text or bounded article subsections with parent-child article mapping; do not cut a legal provision in ways that change meaning.
6. Pass top evidence passages, immutable source IDs and legal-date metadata to LLM; force a structured JSON answer with citations.
7. Validate that each emitted citation ID exists in the retrieved set; if validation fails, return a conservative response or retry once.

9.2 Grounding policy

The model must not answer from its pretrained memory when an answer requires legal authority. It must not invent article numbers, statutory penalties or applicability. System instructions outrank source content: retrieved material is treated as untrusted data and cannot modify tools, policy or output constraints. If facts depend on undisclosed user circumstances, the assistant asks for needed context or states limitations.

9.3 Answer contract

```
{
  "answer_ar": "...",
  "answer_status": "grounded | partial | insufficient_evidence",
  "as_of_date": "YYYY-MM-DD",
  "citations": [
    {"law_id": "...", "article_id": "...",
     "version_id": "...", "source_url": "https://..."}
  ],
  "caveats": ["..."],
  "trace_id": "uuid"
}
```

9.4 Legal response UX

A typical answer presents: (a) a short direct response, (b) legal basis with linked citations and article identifiers, (c) effective-date/status label, (d) a brief informational-use disclaimer and optional suggestion to consult a licensed professional for case-specific advice.

10. Technical architecture

LAYER	RECOMMENDED TECHNOLOGY	RESPONSIBILITY
Frontend	React + TypeScript + Tailwind	Arabic RTL chat, citations, admin dashboard.
API	Python + FastAPI	Request validation, auth, chat and admin endpoints.
Worker / scheduler	Python worker + APScheduler/Cron	Ingestion and scheduled source checks.
Relational & vector DB	PostgreSQL + pgvector	Version metadata, articles, vectors, review events.
Search	PG full-text/trigram + pgvector	Hybrid keyword and semantic retrieval.
LLM / embeddings	OpenAI API (configurable models)	Grounded answer generation and embeddings.
Extraction	PyMuPDF + HTML parser	Text/article extraction and normalization.
Deployment	Docker Compose + HTTPS host	Reproducible local and evaluator deployment.

10.1 High-level component flow



10.2 Suggested repository structure

```
saudi-legal-ai/
  apps/web/                 # React UI
  services/api/             # FastAPI, auth and routes
  services/worker/          # ingestion, feeds, diff, reindex
  packages/legal_core/      # parser, temporal rules, retrieval
  db/migrations/            # tables, indexes and seeds
  tests/unit/
  tests/integration/
  tests/evaluation/
  docs/architecture.md
  docs/source_policy.md
  docker-compose.yml
  .env.example
  README.md
```

11. Data model

ENTITY	KEY FIELDS / RELATIONSHIP
legal_instruments	id; official_title_ar; domain; issuing_authority; instrument_type; canonical_source_url; current_review_status.
document_versions	id; instrument_id; source_url; raw_hash; published_at; effective_from; effective_to; review_status; retrieved_at.
articles	id; version_id; article_number; chapter; original_text_ar; normalized_text_ar; ordering.
article_chunks	id; article_id; chunk_text; embedding; source_offset; lexical_index.
source_registry	id; authority; feed_url; allowlisted_host; enabled; crawl_policy; check_interval.
update_candidates	id; source_id; external_id; discovered_at; changed_hash; target_instrument_id; state.
review_events	id; candidate/version_id; actor_id; decision; decision_at; reason; effective-date evidence.
ingestion_jobs	id; job_type; started_at; finished_at; status; errors; retry_count; counts.
query_traces	id; query_time; as_of_date; retrieved_version_ids; cited_article_ids; response_status; latency_ms.
users	id; role; email/login identity; credential reference; last_login_at.

11.1 Important invariants

- One article belongs to exactly one immutable document version; the current law is a time-dependent view, not a destructively overwritten record.
- Every indexed passage must link to an official source URL, version ID, review state and effective-time interval.
- Only reviewed and legally eligible records appear in default current-law retrieval.
- Re-indexing is idempotent, observable and reversible; deletion of a version from search does not erase audit history.

12. REST API contract and UI requirements

12.1 API endpoint summary

METHOD	PATH	PURPOSE / ACCESS
POST	/api/v1/chat	Question answering; public or authenticated.
GET	/api/v1/laws	Search and filter legal instrument catalog.
GET	/api/v1/laws/{id}/versions	List reviewed versions/effective dates.
POST	/api/v1/admin/documents	Upload document; admin only.
GET	/api/v1/admin/jobs/{id}	Ingestion/job status; admin only.
GET	/api/v1/admin/updates	Update candidates and queue; admin only.
POST	/api/v1/admin/updates/sync	Trigger permitted scan; admin only.
POST	/api/v1/admin/updates/{id}/approve	Validate and approve; admin only.
POST	/api/v1/admin/updates/{id}/reject	Reject with reason; admin only.
POST	/api/v1/admin/versions/{id}/rollback	Rollback index activation; admin only.
GET	/health/live and /health/ready	Operations checks.

12.2 Example user request

```
POST /api/v1/chat
{
  "query": "ما الذي ينص عليه نظام العمل بشأن فترة التجربة؟",
  "domain": "labor",
  "as_of_date": "2026-10-12"
}
```

12.3 Main UI

- Chat screen: message input, optional domain/as-of filters, Arabic RTL rendering, citations beneath substantive claims, state/date banner and open-source action.
- Admin Corpus: document catalog, upload button, parse status, duplicate warnings, version/effective-time timeline.
- Admin Updates: pending changes, official publication URLs, diff view, effective-date evidence, approval/rejection actions.
- Operations: last successful source check, failed jobs, retrieval diagnostics, evaluation metrics and basic audit trail.
- Accessibility: keyboard navigation, readable contrast, logical tab order and RTL controls.

13. Security, privacy, and legal safeguards

CONTROL	MVP REQUIREMENT
Access control	Enforce server-side admin authorization; never rely on hidden UI controls.
Secrets	Store keys in environment/secret store; .env.example only in repository.
Source isolation	Treat retrieved documents as untrusted; prevent instruction injection and HTML/script execution.
Uploads	Constrain MIME/file size, scan content where available; block path traversal and malicious archives.
Personal data	Minimize query logs; redact user identifiers; publish retention/deletion policy.
Data transfer	Assess hosting region and third-party AI data processing under Saudi PDPL rules before real users.
Transport	HTTPS on deployed system; appropriate CORS configuration and rate limiting.
Audit trail	Record review actions and version activation without exposing secrets or sensitive prompts.
User notices	Information service only; citation and effective-date warnings; no attorney-client claim.

Data protection requirements should be checked against the Saudi Personal Data Protection Law and its implementing regulations, including applicable international transfer obligations; the assessment cannot itself certify regulatory compliance. For this MVP, avoid collecting sensitive personal or case data. See the official SDAIA legal resources [S6] and controller guidance [S7].

14. Non-functional requirements

AREA	P0 STANDARD
Latency	Chat p95 ≤8 s in agreed test setup; return meaningful failures rather than indefinite waits.
Reliability	Scheduled monitoring jobs retry transient failures with backoff and expose last-check status.
Data integrity	Duplicate and failed ingestions must not corrupt active article references.
Observability	Structured error logs, request IDs, update-run outcomes and basic monitoring dashboard.
Maintainability	Typed Python/TS, modular pipeline, migrations and clear deployment instructions.

AREA	P0 STANDARD
Reproducibility	Fixture corpus, deterministic test inputs and single-command local startup.
Scalability	MVP optimized for 10–20 documents; architecture supports later bulk import and background workers.
Security	No plaintext credentials in code or logs; authenticated state-changing operations.
Usability	Mobile-readable interface; complete Arabic RTL rendering in user-facing screens.

15. Evaluation plan and acceptance testing

15.1 Test dataset design

CLASS	NUMBER	EXPECTED RESULT
Direct article lookup	15	Correct article, version, citation and short answer.
Paraphrased / Arabic variations	10	Robust retrieval despite wording differences.
Multi-article / cross-document	5	Evidence from all relevant provisions, no invented synthesis.
Out-of-corpus / insufficient evidence	5	Clear abstention and coverage limitation.
Historical and future-effective scenarios	5	Correct time-filtered version and status.
Adversarial / prompt injection	5	Document content cannot override assistant instructions.
Ambiguous or conflicting evidence	5	Qualification, clarification or review flag.
TOTAL	**50**	**Human-graded, repeatable evaluation set.**

15.2 Metrics and grading

- Retrieval Recall@5: fraction of gold-source articles present among top five retrieved passages; target ≥90%.
- Citation accuracy: percentage of cited article references supporting the relevant claim; target ≥95%.
- Grounded correctness: percentage of answerable prompts with all material assertions supported; target ≥90%.
- Abstention on out-of-scope set: target 100% (five deterministic gold tests).
- Effective-date correctness: target 100% on five temporal regression scenarios.
- Operational success: zero broken workflows across upload, candidate detection, approval, re-index and rollback demo.

15.3 Release-blocking demo tests

- Q1: ask a question answerable from the current pilot corpus; inspect article-level citation and open its official URL.
- Q2: ask a question not represented in the corpus; verify abstention and absence of fabricated references.
- Q3: load two time-valid versions of one test article; verify query results differ across as-of dates.
- Q4: trigger a simulated official-feed item twice; verify exactly one update candidate exists.
- Q5: do not approve it; verify the public current answer remains unchanged.
- Q6: approve the fixture amendment with appropriate effective date; verify search/index changes and audit history.
- Q7: inject hostile instructions inside document text; verify source content is not executed as system guidance.
- Q8: disable source access briefly; verify job retries, visible error and no corruption of indexed corpus.

EVIDENCE REQUIREMENT

The evaluation report must show actual measured outputs from the implementation. Do not present proposed thresholds as achieved scores. A reviewer should be able to re-run at least a sample of evaluation questions.

16. Implementation roadmap: 14 working days

DAYS	MILESTONE	DEFINITION OF DONE
1–3	Source registry, schema, ingestion, Arabic parsing	10+ approved documents indexed with original article IDs.
4–6	Hybrid retrieval, answer contract, citations	Answerable questions return verifiable legal references.
7–9	Feed monitor, candidate detection, diff and approval	One approved simulated change updates a versioned index.
10–12	Arabic chat and admin dashboard	Working browser demo for user and reviewer journeys.
13–14	Evaluation, Docker, tests and handover	Hosted demo, 50-case report, clean README and demo video.

16.1 Dependencies and assumptions

- One engineer with working knowledge of Python, FastAPI, SQL, React, Docker and LLM/RAG concepts.
- API/embedding budget, accessible official legal documents, official source review, and an available hosted environment.
- A legally knowledgeable reviewer is available to verify source and effective-date metadata on the demonstration fixture.
- If a source prohibits automated retrieval or requires interactive access, use manual upload and note the limitation in the demo.

16.2 Rough cost categories (not price quotes)

Variable costs are LLM input/output tokens, embedding generation, hosting/database, storage, monitoring, OCR (if needed), and any paid document-access service. Build a usage-based estimate from model token rates and expected questions/day; do not claim a fixed monthly price without load assumptions or verified current vendor pricing.

17. Live demo script and evidence package

17.1 Suggested demo sequence (8–10 minutes)

- 1 minute — problem and source policy: show the selected Saudi official corpus and clarify coverage limits.
- 2 minutes — ask two legal questions, inspect support passage, citation URL, and response metadata.
- 1 minute — ask an out-of-corpus question to demonstrate evidence-based abstention.
- 3 minutes — run a clearly marked simulated amendment through candidate → diff → approval → re-index; compare before and after.
- 1 minute — show previous version and point-in-time answer.
- 1–2 minutes — show evaluation output, API documentation, job logs and Docker launch.

17.2 Submission checklist

- GitHub repository with comprehensible commit history and dependency pinning.
- Working demo URL or reproducible local Docker stack with seeded fictional/test fixture for update simulation.
- README (architecture, setup, environment variables, test execution, known limitations).
- OpenAPI specification and sample requests/responses.
- Corpus manifest listing official sources and approval statuses.

- Evaluation results file with dataset composition, actual accuracy/latency, representative failures and mitigations.
- Short demonstration video and one-page architecture diagram.
- Security and data-handling notes; no API keys, copyrighted bulk dumps or real user records committed.

18. Risks, constraints, and mitigation

RISK	IMPACT	MITIGATION
Incorrect legal effective date	High	Mandatory metadata review; distinct publication/effective fields; temporal tests.
Hallucinated legal citation	High	Citation-ID validation against retrieved evidence; abstention and evaluation.
Feed item not a new law	High	Classify notice types; reviewer gate; source and legal-status verification.
OCR/extraction misses an article	Medium–High	Article-count checks; manual correction; defer scanned PDF automation.
Government website structure changes	Medium	Adapter per source; failure alerts; manual fallback; avoid brittle mass scraping.
Conflicting source versions	High	Preserve provenance and versions; flag conflict and require adjudication.
Cross-border personal-data exposure	High	Synthetic/demo queries; retention minimization; PDPL/transfer assessment.
Narrow-corpus overconfidence	High	Show corpus coverage; filter sources; out-of-corpus refusal.
Schedule overrun	Medium	Prioritize RAG quality and updates; defer P1 UI polish and OCR.

19. Hiring assessment rubric

CAPABILITY	WEIGHT	WHAT A STRONG SUBMISSION DEMONSTRATES
Legal data ingestion and Arabic handling	15	Accurate article parsing, traceable provenance, idempotence.
RAG quality and retrieval	20	Justified hybrid search, version filtering, retrieval metrics.
Grounding, citations and abstention	20	Correct legal citations; defensive behavior on missing evidence.
Monitoring and update lifecycle	20	Traceable official-source watcher, human review, working diff and re-index.
Backend and system design	10	Clean APIs, normalized schema, migrations, error handling.
Testing and evaluation	10	Gold set, measured results, regression and adversarial tests.
UI, deployment and documentation	5	Runnable demo, simple admin flow, readable documentation.
TOTAL	**100**	**Release decision based on demonstrated evidence, not slideware.**

19.1 Suggested evaluation decision

85–100: strong submission; 70–84: promising with clear improvement areas; 55–69: partial demonstration; below 55: key engineering capabilities not proven. These are suggested hiring thresholds for the exercise, not an external standard. Any critical failure involving fabricated legal sources, treating future text as current, or silently auto-publishing unreviewed changes triggers mandatory remediation regardless of numeric score.

20. References and terminology

20.1 Official reference links (verified 12 Oct 2026)

[S1] Bureau of Experts at the Council of Ministers — Saudi Laws Portal

[S2] Umm Al-Qura — Saudi Official Gazette

[S3] Umm Al-Qura — RSS service / Decisions & Regulations feed

[S3a] Direct Decisions & Regulations RSS endpoint

[S4] Ministry of Human Resources & Social Development — Labor Law

[S5] Ministry of Commerce — Regulations

[S6] SDAIA — Personal Data Protection Knowledge Center

[S7] SDAIA — Guide to PDPL for Controllers and Processors

Note: sources are authoritative starting points, not proof of automated ingestion rights or legal coverage. Inspect each source's current terms, relevant publication, and effective-date language during implementation.

20.2 Glossary

TERM	DEFINITION
RAG	Retrieval-Augmented Generation: a model answers using retrieved knowledge passages.
Chunk	A retrievable passage linked to its parent article, version and source.
Hybrid retrieval	Combination of keyword-based and vector similarity search.
Effective date	Date a legal instrument/provision becomes applicable under its legal terms.
Publication date	Date legal text or notice was officially published; not always the date it takes effect.
Version	Immutable snapshot of a legal instrument at a given textual/legal state.
Abstention	Explicitly declining to state a rule when evidence is insufficient.
Source provenance	Traceable origin URL, authority, capture timestamp and content fingerprint.

FINAL DEFINITION OF DONE

The project is complete when an evaluator can ask a supported legal question, open the correct official article, demonstrate a credible reviewed update that changes a time-valid answer, reproduce the evaluation results, and restart the system from the repository.